

Sai Manish Ananthula

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Professional Summary

Results-driven Python Full Stack Developer with hands-on experience building scalable REST APIs, full-stack web applications, and production-grade backend systems using FastAPI, Flask, React.js, and PostgreSQL. Skilled in OOP backend architecture, SQLAlchemy ORM design, authentication and RBAC implementation, and cloud deployments on AWS and DigitalOcean. Proven track record of debugging and stabilizing production systems across frontend, backend, and database layers. Adept at CI/CD automation via Jenkins, containerization with Docker, and delivering clean, maintainable code following SOLID principles.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL

Backend: FastAPI, Flask, Node.js

Frontend: React.js, TypeScript, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Supabase

ORM & Auth: SQLAlchemy, Row Level Security (RLS), RBAC, Supabase Auth, JWT

DevOps & Cloud: Docker, Jenkins, CI/CD, AWS, DigitalOcean, Linux, Nginx

Testing & Tools: Pytest, Postman, REST API Design

Practices: OOP, SOLID Principles, Microservices, DSA, Git, Agile, Debugging, Production Support

Professional Experience

Software Engineer | Bitsilica Pvt. Ltd. | Hyderabad, India

2024 – Present

- Architected and developed object-oriented, modular REST APIs using FastAPI with clean separation of concerns, enabling scalable and maintainable backend services.
- Built full-stack web applications using React.js, TypeScript, FastAPI, SQLAlchemy ORM, and PostgreSQL supporting enterprise workflows.
- Designed and implemented secure authentication systems, role-based access control (RBAC), and backend validation mechanisms to enforce data security policies.
- Engineered optimized relational database schemas and managed SQLAlchemy ORM integrations, improving query efficiency and data integrity across production systems.
- Performed root-cause analysis, log inspection, and production troubleshooting across frontend, backend, database, and cloud environments.
- Developed cron-based automation utilities and backend operational workflows, reducing manual intervention for recurring business processes.
- Containerized applications using Docker and deployed services on AWS and DigitalOcean; integrated Jenkins CI/CD pipelines for automated deployment workflows.
- Implemented Supabase authentication, PostgreSQL storage, and Row Level Security (RLS) policies for granular, policy-driven data access in multi-tenant applications.
- Conducted thorough API testing and debugging using Postman and Pytest, ensuring correctness and performance of backend services in pre-production environments.
- Collaborated with cross-functional teams via Git-based workflows, peer code reviews, and Agile sprint planning to deliver features on schedule.

Key Contributions

Full-Stack Enterprise Platform Development

Technologies: React.js, TypeScript, FastAPI, SQLAlchemy, PostgreSQL, Docker, Jenkins

- Designed and built modular REST APIs with SQLAlchemy ORM, establishing clean data access layers across multiple enterprise-grade applications.
- Managed PostgreSQL schema design and migrations with robust backend validation logic ensuring consistency across multi-user workflows.
- Containerized application stacks with Docker and configured Jenkins CI/CD pipelines for automated, repeatable deployment workflows.

- Resolved production-level API and database bottlenecks through systematic debugging, significantly improving system reliability and uptime.

Role-Based Access Control & Secure Data Architecture

Technologies: React.js, TypeScript, Supabase, PostgreSQL, Row Level Security

- Built role-based access systems with Supabase authentication and fine-grained access control enforced through PostgreSQL Row Level Security (RLS) policies.
- Designed scalable database schemas with audit tracking mechanisms, ensuring full traceability of entity lifecycle events in production environments.
- Resolved complex authentication, data persistence, and access-control issues in production, improving system stability and end-user experience.

Internal Tooling, Automation & Cloud Deployments

Technologies: React.js, FastAPI, MongoDB, SQLite, Jenkins, AWS, Docker

- Integrated FastAPI backend services with React.js frontends to streamline internal operational workflows across teams.
- Configured Jenkins CI/CD pipelines and automated deployment processes; Dockerized services and managed AWS and DigitalOcean deployment configurations.
- Implemented multi-database persistence strategies using MongoDB and SQLite, selecting appropriate storage solutions based on data structure and access patterns.

REST API Development & Backend Services

Technologies: Python, Flask, REST APIs, Postman

- Developed RESTful Flask backend APIs with structured JSON request/response handling and comprehensive input validation.
- Integrated domain logic into modular API services, organizing backend components for long-term maintainability and scalability.
- Performed thorough API testing and edge-case validation using Postman, ensuring correctness across all backend endpoints.

Education

Bachelor of Technology (B.Tech) – Computer Science and Engineering

2020 – 2024

Vardhaman College of Engineering, Hyderabad | CGPA: 7.2

Leadership & Extracurricular

- Volunteered at TEDx, recognized for contribution and coordination activities within the organizing committee.
- Participated in multiple hackathons, collaborating with cross-functional teams to ideate and deliver functional technical solutions under time constraints.
- Actively contributed to debugging, deployment, and production support activities within team environments, demonstrating ownership and reliability.